

Mohammad Haider

Robotics Software Engineer

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SUMMARY

Software engineer with 4 years of experience building correctness-sensitive production systems, telemetry, and analytics infrastructure at Microsoft and the University of Virginia. Now focused on robotics simulation, controls, and measurable experiment design in MuJoCo, with demonstrated work in model-based control, task-space kinematics, support mechanics, and reproducible headless rollouts.

EXPERIENCE

University of Virginia | Data Analyst

May 2025 - Mar 2026

- Owned Tableau analytics products for institutional stakeholders from discovery through implementation, deployment, and ongoing support.
- Translated ambiguous operational questions into SQL-backed metric definitions and decision-support dashboards.
- Collaborated with colleagues to investigate Tableau performance bottlenecks using telemetry, informing optimization work that improved organization-wide dashboard load times by approximately 20%.

Microsoft | Software Engineer - Data & Analytics

Jul 2021 - Jul 2024

- Migrated Microsoft's internal financial reporting platform from SSAS Multidimensional to Tabular with zero downtime, owning implementation, telemetry, production validation, and operational support.
- Extended automated financial-value reconciliation between legacy and replacement systems, preserving reporting correctness throughout the migration and improving data quality.
- Reduced representative filtering latency by up to 200x and user-visible Excel loading time by approximately 4x, measured through production telemetry and user reports.
- Built and maintained Azure-based financial reporting pipelines supporting business-critical planning and financial close operations.

Microsoft | Software Engineering Intern

Summers 2017, 2018

- Built internal engineering tools supporting reporting infrastructure and developer workflows across multiple engineering teams.

SELECTED PROJECTS

C-1N + Robotics Test Bench | Python, MuJoCo

- Built a six-legged MuJoCo robot simulation with 12 actuated joints, torso-orientation and foot-contact telemetry, and interactive and headless rollout modes; preserved the current non-sustained tripod shuffle as an explicit failure baseline rather than presenting it as walking.
- Built controlled simulation experiments around joint coupling, model-based control, Jacobians, and static support, changing one meaningful condition at a time and recording predictions before execution.
- Measured matched-model computed torque at approximately 0.0004 and 0.0005 rad per-joint RMS position error versus approximately 0.68 and 0.08 rad under PD, and verified MuJoCo task-space Jacobians against finite differences and foot velocities at numerical precision.

Indigo Circuit | Flask, DuckDB, dbt

- Built and operated a Railway-hosted analytics platform with nightly idempotent ingestion and fail-closed shadow-database validation before atomic production swaps.

TECHNICAL SKILLS

Languages: Python, C#, SQL, Java, R

Robotics & simulation: MuJoCo, model-based control, inverse dynamics, Jacobians, task-space kinematics, simulation telemetry

Data platforms: DuckDB, dbt, Azure Data Factory, Azure Data Lake, SSAS Tabular

Engineering: Flask, PostgreSQL, REST APIs, Git, Azure, monitoring, validation, performance optimization

EDUCATION

Virginia Tech | B.S. Statistics